



## Review

# Modernising meat inspection of pigs – A review of the Danish process from 2006–2020

Lis Alban<sup>a,b,\*</sup>, Jesper Valentin Petersen<sup>a</sup>, Anne Kristine Bækbo<sup>a</sup>, Tanja Østergaard Pedersen<sup>c</sup>, Amanda Brinch Kruse<sup>b</sup>, Goncalo Pacheco<sup>d</sup>, Marianne Halberg Larsen<sup>b</sup>

<sup>a</sup> Danish Agriculture & Food Council, Axeltorv 3, DK-1609, Copenhagen V, Denmark

<sup>b</sup> Department of Veterinary and Animal Sciences, Faculty of Health and Medical Sciences, University of Copenhagen, Grønnegårdsvej 8, DK-1870, Frederiksberg C, Denmark

<sup>c</sup> Eurofins Biopharma Product Testing A/S, Ørnebjergvej 1, DK-2600, Glostrup, Denmark

<sup>d</sup> Danish Veterinary and Food Administration, Stationsparken 31-33, DK-2600, Glostrup, Denmark

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## ABSTRACT

In 2004, the European Union legislation allowed to change meat inspection of finishing pigs, raised under controlled housing conditions, pending use of risk assessment of the effect of the proposed changes. This paper summarises the process undertaken in Denmark, where a series of national risk assessments and epidemiological investigations were undertaken, in a stepwise approach to address specific needs due to trade requirements.

The initial focus was to examine the visual-only inspection (VOI) in lieu of the traditional inspection method, which involves incisions and palpations of various organs. The objective was to understand what would be missed, how often, and what the consequences might be, if incisions and palpations would no longer be undertaken routinely. Next, the microbiological burden of pigs previously suffering from generalised diseases was evaluated.

The work was undertaken in collaboration among the academia, industry, and veterinary authorities. The OIE (Office International des Epizooties, now World Organisation for Animal Health) approach to risk assessment was used, which involves steps of hazard identification, release assessment, exposure assessment, consequence assessment, and risk estimation. To inform the results of assessment, samples were collected from abattoirs and subjected to laboratory investigation. Furthermore, slaughterhouse statistics, literature, and expert opinion were used.

It was concluded that VOI could safely replace traditional inspection, although some cases of endocarditis and embolic pneumonia would be missed. Today, VOI is in place for finishing pigs that are raised indoors. The information collected in this study has resulted in an update of the mandatory septicaemia-inspection at the rework area. Moreover, the Danish legislation has been updated; replacing mandatory de-boning of prior septicaemia cases with an own-check program involving targeted focus on predilection sites for abscesses during cutting.

## 1. Introduction

Worldwide, veterinary authorities are facing a challenge related to surveillance, inspection, and control of food safety, animal health and animal welfare, principally due to scarce resources and plentiful challenges. Therefore, cost-effective ways are sought for, developing a valuation of the individual parts of a given surveillance program. Moreover, it should be questioned whether the actions taken are fulfilling the objectives for which they were originally intended.

Meat inspection is an example of a surveillance portfolio, covering many different aspects to secure food safety, animal health, and welfare. The core of meat inspection was defined more than one hundred years ago, and it is based upon lesions, which can be seen macroscopically at the time of inspection. It is necessary to re-evaluate at regular intervals whether inspection is still targeting the hazards that make people or animals ill (Mousing et al., 1997). This had been considered by the regulatory authorities already two decades ago when a white paper for food safety as well as an Opinion of revision of meat inspection

\* Corresponding author. Danish Agriculture & Food Council, Axeltorv 3, DK-1609, Copenhagen V, Denmark.

E-mail address: [lia@lf.dk](mailto:lia@lf.dk) (L. Alban).

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procedures were published, pointing to the needs of reviewing certain meat inspection requirements as they were no longer considered fit-for-purpose (Anonymous, 2000a; Anonymous, 2000b). The intention was to change inspection to be risk-based and visual-only, while focusing on the entire production chain and keeping in mind the responsibility of the food business operators (Anonymous, 2000a; Anonymous, 2000b). These concepts were subsequently incorporated into the General Food Law of the European Union (EU) (Anonymous, 2002).

Several of the cuts and incisions which form part of the traditional inspection are targeting infections which may no longer be relevant in a country, a region, or a given animal species. For example, *Mycobacterium bovis* in pigs typically results in lymphadenitis in the mandibular lymph nodes, and therefore, mandibular lymph nodes are incised as part of traditional meat inspection (Alban et al., 2008). However, several European countries have been free from bovine tuberculosis for decades. If bovine tuberculosis is reintroduced, then it would occur in imported cattle or llamas. Wildlife may be infected in specific geographical areas, and here a spill-over to domestic animals, such as bovines or outdoor-reared pigs, may be seen occasionally. In contrast, indoor-reared pigs are less likely to become exposed and infected. In this case, a risk-based surveillance programme in selected cattle herds and outdoor pig herds may be more cost-effective than looking at all animals (Hansen et al., 2018).

The main pork-related microbiological foodborne hazards identified by the European Food Safety Authority (EFSA) include *Salmonella* spp., *Yersinia enterocolitica*, *Toxoplasma gondii*, and *Trichinella* spp. (EFSA, 2011a). These hazards cannot be identified through traditional meat inspection as they are not necessarily associated with macroscopically visible lesions (Buncic et al., 2019). Detection of a lesion during inspection may not in itself represent a food safety issue, as most lesions observed are not reflecting an acute stage. Moreover, no or very few pathogenic bacteria may be present in the observed lesions, or the lesion is present in/on non-edible tissue. Finally, many microbiological agents capable of causing disease in pigs are not zoonotic (Alban et al., 2008; Mousing et al., 1997). In fact, to elucidate the impact of a lesion on human health, more focus should be on the microbiological aspects.

Modernisation of meat inspection is therefore on the agenda in several countries. In 2004, the EU Meat Inspection Regulation 854/2004 introduced the possibility to modify the traditional inspection of swine, pending on two conditions: i) that it was only applicable for finishing pigs from controlled housing, implying herds with a high level of biosecurity, and ii) a risk assessment should confirm that the proposed change would ensure a level of protection at least equal to traditional inspection (Anonymous, 2004). This initiated a modernisation process in several EU Member states. An update of the challenges and the status of modernised meat inspection can be found in Alban et al. (2018), Buncic et al. (2019), Bækbo et al. (2015), and Riess & Hoelzer, (under revision).

Pork is traded internationally, and some countries are dependent upon export. Denmark is an example of this as it exports most of its pork with an annual value of approximately €4 billion (Danish Agriculture & Food Council, 2019). Before an exporting country can introduce changes to its meat inspection, acceptance of equivalence is needed. This means that an exporting country should be able to substantiate that the suggested changes would not jeopardise food safety or ability to detect important animal infections. Moreover, it should be documented that the new or alternative way of performing meat inspection can be considered as equivalent to the importing country's inspection. Finally, the documentation should be transparent and science-based. For this, a risk assessment approach is recommended by the OIE (Office International des Epizooties, now World Organisation for Animal Health (OIE, 2010)).

At the early beginning, two important studies were undertaken in Denmark. The first dealt with the consequences of a change from a traditional to visual-only inspection (VOI). This was assessed in a comparative study in 183,383 slaughter pigs in 1993. The results

showed that only a few cases of importance for food safety would remain undetected as a result of changing from traditional to visual inspection (Mousing et al., 1997). The reasons for implementing VOI were the potential for decreased cross-contamination as a result of no handling, cutting, or incisions as well as reduced inspection costs, allowing reallocation of the resources saved to hygiene and surveillance programmes (Mousing et al., 1997).

The second study involved more than 3 million pigs and was undertaken in 1996. It dealt with an evaluation of the traditional inspection procedures for pig heads on the slaughter line and their impact on food safety (Sørensen & Petersen, 1999). Moreover, Petersen et al. (2002) argued that a modified slaughter and inspection technique omitting palpation and incision would decrease the contamination of the carcass with pathogenic microorganisms, and that lesions in the head left undiscovered would be of little or no importance either for human health or for the overall supervision of animal health.

The opening in the EU Regulation 854/2004 for changes to the meat inspection regime resulted in a series of studies, undertaken in Denmark from 2006 to 2020. In the following, we shall explain how the risk assessment approach was used in these studies to guide the modernisation process. Because of these studies, a gradual move was made from the traditional inspection to visual-only, and subsequently to developing new ways of handling of pigs with lesions indicative of prior septicaemia, while focusing on minimising food waste without compromising food safety.

## 2. Materials and methods

In line with the EU Meat Inspection Regulation from 2004, we decided to undertake a risk assessment for finishing pigs raised under controlled housing conditions. This means an animal husbandry type model, where swine are raised under housing and feeding conditions which are controlled by the food business operators. The requirements for a herd to be considered as controlled housing were initially specified in the now former EU *Trichinella* Regulation from 2005 (Anonymous, 2005), today listed in Annex IV of the current *Trichinella* Regulation (Anonymous, 2015). Most of the pigs raised in Denmark belong to the controlled housing model, as only 5% of the finishing pigs are raised under non-controlled conditions including free-range production (Hansen et al., 2018).

The OIE approach to risk assessment was chosen, which involves steps of hazard identification, release assessment, exposure assessment, consequence assessment, and risk estimation (OIE, 2010). Although some parts were quantitative, a qualitative approach was mostly used to sum up the risk. To inform the various elements of the risk assessment, in-country-up-to-date data were collected in the form of samples from organs, muscle, and lesions from abattoirs. These samples were subjected to relevant laboratory investigations including Matrix-assisted laser desorption-time of flight mass spectrometry (MALDI-TOF MS). Furthermore, slaughterhouse statistics, literature, and expert opinion were used. Based upon the collected data, simple statistics were calculated and more advanced statistical methods such as principal component analysis were used, depending on the exact question. The year of the individual study, objective, hazards identified, study design, number of plants and samples involved, laboratory work undertaken, and statistics are described for each study in the next section. For additional details, please see the individual papers and reports. Based upon this, the probability of release, exposure, and consequences were summarised into a risk estimate. Moreover, the intended use of the meat and the conclusion of the study are presented. Furthermore, it is documented if the study resulted in a change of the official handling or the legislation.

The process was stepwise, allowing a gradual implementation of the changes. The entire process is depicted in Fig. 1. The first three assessments dealt with what would be missed, how often, and what the consequences might be, if routine incisions and palpations into the heart, the mandibular and the gastro-intestinal lymph nodes, and the lungs

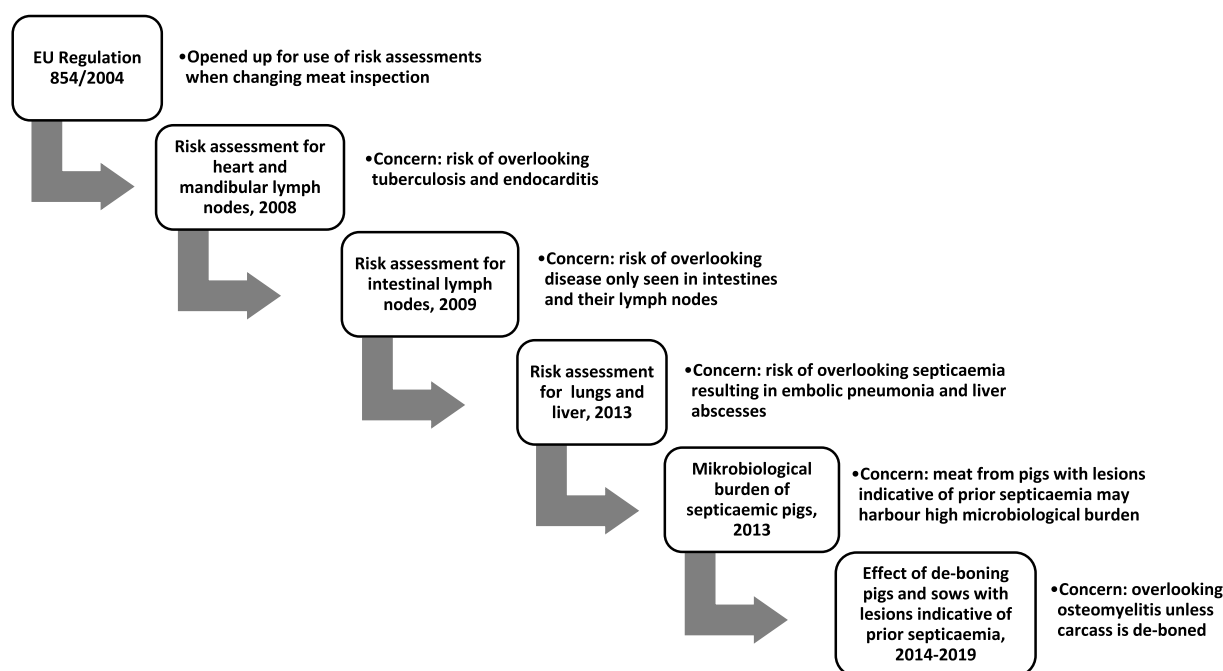


Fig. 1. Description of the modernisation of meat inspection process for swine in Denmark.

were to no longer be undertaken.

Through the risk assessment for the lungs, it became clear that cases of embolic pneumonia might be missed if lungs are not palpated routinely and lesions are not detected in other organs. Therefore, it was decided to undertake a specific study of finishing pigs with embolic pneumonia to learn what the microbiological burden of such pigs was.

Next, focus was moved to pigs with purulent lesions, indicative of prior septicaemia. These lesions are caused by a traumatic injury such as a tail bite, occurring months earlier. According to Danish legislation at that time, acute cases were condemned, whereas chronic cases were subjected to mandatory de-boning (Anonymous, 2011). Mandatory de-boning means a separation of all muscles from bones of the carcass, followed by a thorough visual inspection of the bones for lesions. Subsequently, the bones would be designated to pressure sterilisation, irrespective of lesions detected or not, due to the allocation of this animal by-product into Category 2, as described in the EU Regulation 1069/2009. Category 1 animal by-products require incineration, Category 2 pressure sterilisation, whereas Category 3 can in principle be used raw although it is often heat-treated (Anonymous, 2009). The aim of de-boning was to ensure detection of osteomyelitis, not found during the mandatory septicaemia-inspection. The question was whether de-boning was needed to ensure food safety and whether another handling would be equally safe.

The Danish Agriculture & Food Council acted as the main driver in the process, in collaboration with the University of Copenhagen and in a dialogue with the Danish Veterinary and Food Administration. Most risk assessments were subjected to external peer reviews. Such reviews were made either by the Danish Technical University, independent researchers, or administrative staff affiliated with the importing country's veterinary authorities. Moreover, all risk assessments have been published, some in peer reviewed journals other as conference proceedings or on the home page of Danish Agriculture & Food Council (<https://lf.dk/aktuelt/publikationer/svinekod>).

### 3. Results

#### 3.1. Risk assessment for the heart

This risk assessment was undertaken in 2008. The objective was to

study what would be missed if the heart was no longer opened routinely as a part of meat inspection (Alban et al., 2008). Parasitic lesions were ruled out, because *Taenia solium* is not present in Denmark, and *Ascaris suum* results mainly in lesions observed on the liver (milk spots) and in the lungs. Abscesses as part of the septicaemia complex have predilection sites, and the myocardium is not among these sites. Finally, lesions related to pericarditis can be observed from the outer side of the heart. Endocarditis was identified as a lesion, which would be missed, because detection requires opening of the heart. Endocarditis is seen on the valves of the heart (Jensen et al., 2006). The most important hazards related to endocarditis in pigs, identified in the literature, were *Streptococcus suis* and *Erysipelothrix rhusiopathiae* (Jensen et al., 2006).

According to Danish meat inspection data covering the time period from 1999 to 2007, the prevalence of endocarditis in finishing pigs was slightly lower than 0.01%. Hence, the condition occurs infrequently. In 2008, a case control study was undertaken involving 88 hearts with visually recognisable endocarditis and 57 hearts without visually recognisable endocarditis, collected from finishing pigs delivered for slaughter at 10 different plants in Denmark. The hearts were subjected to standard microbiological analyses. Among the hearts with endocarditis, *S. suis* and *E. rhusiopathiae* were found in 77%, whereas 79% of the hearts without endocarditis were culture-negative (Table 1). Hence, the laboratory results supported the findings from the literature. Both *S. suis* and *E. rhusiopathiae* are considered occupational hazards, implying that they may cause infection in already existing wounds (Alban et al., 2018). A contact to the local union of slaughterhouse workers revealed that such cases were not reported – probably because of many years of focus on ensuring safe working conditions (Alban et al., 2008). If destined for human consumption, hearts are usually cooked for more than an hour. However, shorter cooking times of sliced pork heart is also possible. Both kinds of treatment are expected to eliminate all bacteria on the surface of the heart.

The conclusion was that the routine opening of the heart from apparently healthy finishing pigs could be omitted as part of the meat inspection without any risk for food safety, animal health, or animal welfare. However, if other lesions were found indicating septicaemia, then the carcass and plucks should go to the rework area for an extended examination as part of the septicaemia-inspection, whereas the gut and intestines would be condemned. Today, to lower exposure of consumers

**Table 1**

Distribution of microorganisms found in a study of 88 pig hearts with endocarditis and 57 pig heart without endocarditis found at the slaughterline, Denmark 2008, modified after Alban et al. (2008).

| Microorganism                           | Number of samples (%) |                      |
|-----------------------------------------|-----------------------|----------------------|
|                                         | With endocarditis     | Without endocarditis |
| <i>Streptococcus suis</i> like          | 40 (45.5)             |                      |
| <i>Erysipelothrix rhusiopathiae</i>     | 28 (31.8)             |                      |
| Beta-hemolytic streptococci             | 5 (5.7)               |                      |
| Mixed culture with <i>Streptococcus</i> | 0 (0.0)               | 2 (3.5)              |
| <i>Lactobacillus garvieae</i>           | 4 (4.5)               |                      |
| <i>Proteus</i>                          | 0 (0.0)               | 1 (1.8)              |
| <i>Trueperella pyogenes</i>             | 1 (1.1)               |                      |
| Isolates not further identified         | 5 (5.7)               |                      |
| Culture-negative                        | 5 (5.7)               | 45 (78.9)            |
| Contaminated                            | 0 (0.0)               | 9 (15.8)             |
| Total                                   | 88 (100.0)            | 57 (100.0)           |

to *S. suis* and *E. rhusiopathiae*, the hearts are opened by an abattoir worker, instructed specifically in condemning hearts with lesions indicative of endocarditis. No further actions are taken in case endocarditis is seen as the only lesion observed on a carcass (Alban et al., 2008).

### 3.2. Risk assessment for the mandibular and the gastro-intestinal lymph nodes

This risk assessment was divided in two parts; the first part dealt with the mandibular lymph nodes and was undertaken in 2008 (Alban et al., 2008). The second part dealt with the gastro-intestinal lymph nodes and was undertaken in 2009 and published in English in 2010 (Alban et al., 2010). The objective was to study what would be missed if the mandibular and the gastro-intestinal lymph nodes were not incised routinely. The most important hazard identified through the literature was *Mycobacterium bovis*. Infection with *M. bovis* might result in development of granulomatous lesions in the lymph nodes. Often necrosis is present (caseous lymphadenitis) and/or mineralization (Jensen et al., 2006). *M. avium* may also be found in lymph nodes with granulomatous lesions (Hansen et al., 2018; Komijn et al., 2007). Other relevant hazards include *Rhodococcus equi*, neoplasms, fungi, and parasites as a result of visceral larvae migrans (Jensen et al., 2006).

According to Danish meat inspection data covering the time period from 1999 to 2007, lymphadenitis was observed infrequently, with a prevalence ranging from 0.01% to 0.02% (Alban et al., 2008). In 2008, a case study was undertaken involving 43 mandibular lymph nodes with granulomatous lesions and/or caseous necrosis collected from finishing pigs delivered for slaughter in Denmark. The samples were subjected to Ziehl-Neelsen staining and standard microbiological analyses. A total of 63% of these samples were positive for *Rhodococcus equi* and all were negative for mycobacteria (Table 2). Hence, the laboratory results supported the findings from the literature (Alban et al., 2008).

None of the identified hazards are considered meat-borne. *M. bovis* is a zoonosis, where consumption of raw, infected milk and close contact to infected animals may lead to human infection, whereas the prevailing

**Table 2**

Distribution of different microorganisms found in a study of 43 mandibular lymph nodes with granulomatous lesions and or caseous necrosis in Danish finisher pigs, 2008, modified after Alban et al. (2008).

| Microorganism                          | Number of samples (%) |
|----------------------------------------|-----------------------|
| Negative for <i>Mycobacterium</i> spp. | 43 (100.0)            |
| <i>Rhodococcus equi</i>                | 27 (62.8)             |
| <i>Nocardia farcinica</i>              | 1 (2.3)               |
| Culture-negative                       | 15 (34.9)             |
| Total                                  | 43 (100.0)            |

opinion is that consumption of meat represents a minute risk (Alban et al., 2008). *M. bovis* is also an animal health hazard. Denmark is officially free from *M. bovis* since 1980, and a surveillance programme is in place. Denmark only imports a very low number of pigs and a low number of cattle annually, and requirements for testing and quarantine are in place. Hence, the import risk for *M. bovis* is judged as low (Foddai et al., 2015). Bovine tuberculosis has been found in farmed deer in Denmark previously, and the last case was observed in 1988 (Anonymus, 2019a). *M. avium* may be a differential diagnosis for *M. bovis*. Occasionally, outbreaks of *M. avium* have been observed in pigs, usually related to use of raw peat. Since 2017, use of peat for pigs is prohibited in Danish pig production (Hansen et al., 2018).

The mandibular and gastro-intestinal lymph nodes are not considered as edible tissue; the tissue is cut away when cleansing the neck and used as pet food after heat treatment at the plants processing animal by-product categories two and three. The conclusion was that the routine incisions and/or palpations of these lymph nodes could be omitted without jeopardising food safety, animal health, or welfare. If one of these lymph nodes appears to be swollen or otherwise non-normal, then further examination is warranted (Alban et al., 2008, 2010). The studies presented in Section 3.1 and 3.2 led to introduction of a partial visual inspection - implying no routine incision into the mandibular lymph nodes and no routine opening of the heart on otherwise healthy finisher pigs, raised under controlled housing conditions after acceptance of equivalence by trade partners in 2008.

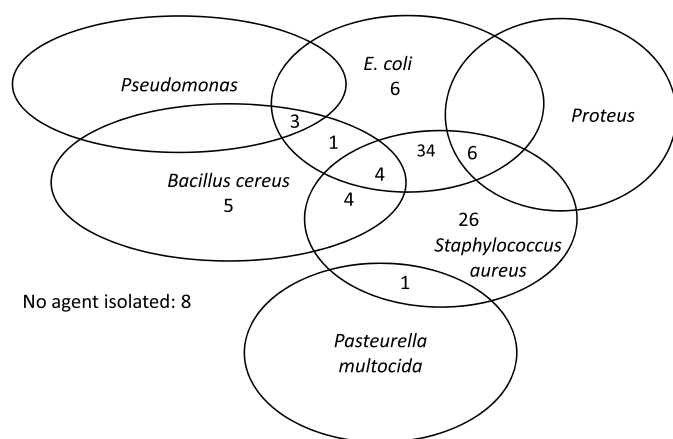
### 3.3. Risk assessment for the lungs

This risk assessment was undertaken in 2013 and the details can be found in Pacheco et al. (2013). The objective was to study what would happen if the lungs were not palpated routinely. Many lung lesions manifest themselves with macroscopically detectable lesions, and hence, will not escape visual detection. However, embolic pneumonia was identified as a condition, which may be overlooked if the emboli are small or located deeper in the tissue. In 2012, a case study was undertaken involving 104 lungs with embolic pneumonia and 98 associated lymph nodes collected in relation to slaughter and inspection of indoor-raised finishers at three different Danish abattoirs. The samples were subjected to standard microbiological analyses. The lung results depicted in Fig. 2 show that *Staphylococcus* spp. was the most commonly found microorganism followed by *Escherichia coli*, and *Bacillus cereus* (Pacheco et al., 2013). Although lesions found at meat inspection are routinely registered into a database, there is no specific code for embolic pneumonia. The code "blood poisoning" is used for any septicaemia- or pyaemia-related lesions such as embolic liver abscesses, embolic nephritis, splenitis, and embolic pneumonia. In general, presence of such lesions results in a requirement for total condemnation, unless the lesions are judged as non-acute (in Danish called chronic). In that case, the final decision regarding the carcass was pending the de-boning findings, which could lead to total condemnation or acceptance after partial condemnation (Pacheco et al., 2013).

Data were retrieved from the meat inspection database for the largest abattoir plant in Denmark, covering 1.2 million finishing pigs slaughtered between October and December 2012. These data showed that the prevalence of the lesion code "blood poisoning" was 0.06% in finishing pigs, implying that the condition was uncommon. Moreover, "blood poisoning" was identified as a risk factor for presence of abscesses with a relative risk (RR) of 4.4 ( $P < 0.0001$ ). However, most abscesses (97%) were found in pigs without lesions indicative of "blood poisoning" (Table 3).

A comparison between VOI and traditional inspection of plucks of finishing pigs was undertaken by two experienced meat inspectors (where Observer 1 was one of the authors) 2 days per week, during 3 weeks in October 2012, on one abattoir, enabling inspection of 3000 plucks (Table 4; Pacheco et al., 2013). In general, the agreement between observers, measured as kappa (ranging from -1 to 1 where 0 = no





**Fig. 2.** Graphical description of main part of bacteriological findings in 104 lungs with lesions indicative of embolic pneumonia from Danish finisher pigs, raised indoors. Modified after Pacheco et al. (2013). Due to lack of space, bacteria found in seven lung samples are not displayed in the figure: *B. cereus*, *E. coli*, *S. aureus*, *P. multocida*: N = 2. *E. coli*, *S. aureus* and *Pseudomonas*: N = 2. *B. cereus*, *Pseudomonas* and *S. aureus*: N = 1. *B. cereus*, *E. coli*, *Pseudomonas* and *S. aureus*: N = 2.

**Table 3**

Association between presence of lesions indicative of pyaemia and presence of abscesses found in finisher pigs, slaughtered at Danish Crown Horsens, October–December 2012. Modified after Pacheco et al. (2013).

| Presence of lesions indicative of pyaemia | Presence of abscess (es) |           | Assessment of association        |
|-------------------------------------------|--------------------------|-----------|----------------------------------|
|                                           | Yes                      | No        |                                  |
| Yes                                       | 97                       | 620       | Relative risk = 4.4              |
| No                                        | 37,195                   | 1,172,660 | P-value < 0.0001                 |
| Sum                                       | 37,292                   | 1,173,280 | Positive predictive value = 0.14 |

agreement beyond chance, and 1 = full agreement), was low. Observer 1 undertaking VOI found more lesions than Observer 2 undertaking routine traditional inspection as required by the Danish authorities. This occurred because Observer 1 was asked to note all lesions, whereas Observer 2 conducted the ordinary meat inspection, and therefore, only registered lesions requiring action by the meat inspection team. Among the nine lesions codes used for the plucks of the 3000 pigs, only heart lesions were associated with a high agreement between observers (kappa = 0.71).

In 2012, a scoring exercise for lungs with septicaemia (N = 104), detected by the meat inspectors, was undertaken by two persons on 1 day in one plant. One of the two persons had experience with practical meat inspection and the other not (two of the authors). The two persons assessed together whether they thought they would have been able to detect the case if they had not known that the meat inspectors had judged the lungs as harbouring emboli, based on the traditional inspection consisting of a combination of palpation and visual inspection. Although such an assessment is far from objective, it indicated to that around 63% of these lesions were assessed as being visually detectable, whereas the answer was “maybe” for 18% and “no” for 19%. Hence, as a best guess, between one fifth and one third of the cases would be missed if visual-only were to replace traditional inspection of lungs (Pacheco et al., 2013).

The conclusion of the study was that although presence of embolic pneumonia may be missed when undertaking VOI, routine palpation of the lungs can be omitted, because the bacteria involved are usually not zoonotic and if they are, they are not meat-borne. In Denmark, lungs were (and still are) not considered edible tissue but designated as

**Table 4**

Comparison of findings in visual-only versus traditional meat inspection of 3000 Danish finisher pigs raised indoors, 2012 (modified after Pacheco et al., 2013).

| No. of carcasses found with lesions during |                        | Assessment of association <sup>a</sup> |                      |       |
|--------------------------------------------|------------------------|----------------------------------------|----------------------|-------|
| Visual inspection                          | Traditional inspection | Visual <sup>b</sup> versus traditional | P-value <sup>b</sup> | Kappa |
| Any lesion                                 | + –                    |                                        |                      |       |
|                                            | 232 338                | 1.6                                    | <0.0001              | 0.42  |
|                                            | 117 2313               |                                        |                      |       |
| Acute pleuritis                            | + –                    |                                        |                      |       |
|                                            | 0 1                    | N.A.                                   | N.A.                 | N.A.  |
|                                            | 0 2999                 |                                        |                      |       |
| Acute pleuropneumonia                      | + –                    |                                        |                      |       |
|                                            | 4 48                   | 13.0                                   | <0.0001              | 0.14  |
|                                            | 0 2948                 |                                        |                      |       |
| Catharal bronchopneumonia                  | + –                    |                                        |                      |       |
|                                            | 2 317                  | 159.5                                  | <0.0001              | 0.01  |
|                                            | 0 2680                 |                                        |                      |       |
| Contamination                              | + –                    |                                        |                      |       |
|                                            | 2 38                   | 13.3                                   | <0.0001              | 0.09  |
|                                            | 1 2959                 |                                        |                      |       |
| Embolic pneumonia                          | + –                    |                                        |                      |       |
|                                            | 2 7                    | 3.0                                    | 0.0339               | 0.33  |
|                                            | 1 2990                 |                                        |                      |       |
| Heart lesions                              | + –                    |                                        |                      |       |
|                                            | 110 57                 | 1.2                                    | 0.0017               | 0.71  |
|                                            | 28 2805                |                                        |                      |       |
| Kidney lesions                             | + –                    |                                        |                      |       |
|                                            | 29 60                  | 0.68                                   | 0.001                | 0.24  |
|                                            | 102 2809               |                                        |                      |       |
| Liver lesions                              | + –                    |                                        |                      |       |
|                                            | 10 70                  | 3.2                                    | <0.0001              | 0.18  |
|                                            | 15 2905                |                                        |                      |       |
| Other                                      | + –                    |                                        |                      |       |
|                                            | 1 1                    | 0.01                                   | <0.0001              | 0.02  |
|                                            | 90 2908                |                                        |                      |       |

N.A.: Not applicable.

<sup>a</sup> The prevalence found during visual inspection divided with the prevalence found at traditional inspection.

<sup>b</sup> McNemar's Chi square test for matched pair of subjects. P-value < 0.05 indicates that the proportions are statistically different.

Category 3 animal by-products. In most cases, the lungs were heat-treated before being used as pet food or feed for fur animals, although use as “bones and raw feed” (BARF) also took place to some and unknown extent. If the lungs had a non-normal appearance or lesions were found in other organs are indicating septicaemia, then further examination was judged as warranted. Moreover, such lungs were designated to Category 2 animal by-products, and hence subjected to pressure sterilisation. This study led to implementation of VOI of finisher pigs raised under controlled housing conditions after acceptance of equivalence by trade partners in 2015.

### 3.4. Assessment of the microbiological burden of pigs with lesions indicative of septicaemia

The risk assessment for omitting palpation of lungs had shown that the most commonly found bacteria in the lungs of pigs with embolic pneumonia were *S. aureus* and *E. coli* (Fig. 2; Pacheco et al., 2013). The next question was to elucidate whether the bacteria causing the embolic pneumonia also could be found in the muscles or other tissues intended for human consumption. Hereby, the implication for food safety related to missing embolic pneumonia could be assessed.

In 2013, a case control study was undertaken and included 19 finishing pigs with embolic pneumonia, identified by the meat inspectors in one plant (Kruse et al., 2015). For each pig, samples were taken from the lungs, liver, heart, kidney, spleen, a joint, and a muscle (distal on the right foreleg). Moreover, 60 muscle samples, also distal on the right

foreleg, were collected from healthy pigs from the same abattoir. These samples were subjected to standard microbiological analyses including bacterial identification using MALDI-TOF MS (Kruse et al., 2015).

Bacteria were isolated from 78 out of 127 tissue and swab samples corresponding to 61% positive samples. *S. aureus* was the most frequently isolated bacterium with 37 positive samples, and the predominant site was the lungs. One of the 37 isolates was identified as methicillin-resistant *S. aureus* (MRSA). *S. warneri* was the second-most commonly found bacterium (N = 24), with less than 50 CFU/g in the positive samples and without any predominant site. This was followed by *E. coli* (N = 8) and *Micrococcus luteus* (N = 6) as well as other less-frequently found bacteria. The control group consisting of 60 pigs revealed one positive sample for *S. aureus* and the relative risk of finding *S. aureus* was 20 times higher in the case group compared to the control group (P = 0.0003). In summary, the laboratory analysis showed that bacteria were only seldom present in muscles, and if they were, the numbers were low (<10 CFU/g). The interpretation was that the animal had in most cases cleared itself from the original septicaemia, which occurred maybe months earlier (Kruse et al., 2015).

The bacteria detected were not known as food-borne hazards, but some of them - such as staphylococci - may be considered as occupational hazards and result in soft tissue infections. Toxins related to staphylococci may cause food-borne illness. However, according to EFSA, the risk of human disease does not seem to be correlated with occurrence of *S. aureus* in raw pork but rather to improper food handling and storage enabling growth-related toxin production (EFSA, 2011). In line, EFSA did not consider *S. aureus* as a relevant hazard for the meat inspection of pigs (EFSA, 2011).

The meat inspectors sent 16 of the 19 pigs to de-boning due to the local, non-acute nature of the lesions found, and the remaining three were condemned because of presence of lesions judged as of acute character. There was no association between the microbiological findings and the meat inspectors' decision; the three condemned cases had fewer samples positive for *S. aureus* than the average of the 16 other animals (Kruse et al., 2015). As stated above, pig lungs are not destined for human consumption in Denmark but for feed for fur animals or pets, mainly after sufficient heat treatment. The conclusion was that there was no increased risk for human health associated with consumption of meat from a pig, where embolic lesion indicative of prior septicaemia had been missed.

### 3.5. Assessment of value of de-boning of pigs and sows with lesions indicative of prior septicaemia

As shown in the studies described above, abscesses may be found during meat inspection, indicating that an animal has suffered from septicaemia at an earlier stage in life. According to Danish legislation at that time, such animals were to be sent to the rework area for an extended septicaemia-inspection with incisions targeting the predilection sites for such abscesses. Tissue with abscesses would be removed. Next, the carcass would be sent for mandatory de-boning because of a probability of harboring osteomyelitis not found during inspection. After de-boning, most carcasses would be accepted. However, all bones separated from the de-boned carcasses would be condemned and treated as category 2 animal by-products. The food safety value of this approach had been questioned, because abscesses are regarded as a quality issue with no impact on food safety (Bækbo et al., 2016).

To study this, an investigation was undertaken in finishing pigs in 2013 (Bækbo et al., 2016) and in 2015 for sows (Pedersen et al., 2017). The objective was to 1) map the location of abscesses found during the mandatory septicaemia-inspection in the rework area, 2) assess the proportion of pigs where additional abscesses were found during de-boning, 3) study association between presence of bacteria in the abscess and the muscle meat to assess the food safety impact of such abscesses.

Two case-control studies were undertaken involving one abattoir

slaughtering finishing pigs and one abattoir slaughtering sows. The studies involved samples from 102 finishing pigs and 105 sows with purulent lesions destined for de-boning as well as 60 finishing pigs and 60 sows, fully approved at meat inspection. For all cases, samples were taken from abscesses and the muscle (distal on the right foreleg). For the controls, only a muscle sample was taken, also distal on the right foreleg. All samples were subjected to MALDI-TOF MS for bacterial identification. In finishing pigs, the most common location of abscesses found was the thoracic cavity (42%), whereas it was the tenderloin for sows (32%).

In finishing pigs, the bacterial concentration was below the level of detection in 83% of the muscle samples, whereas in the abscesses, it was only 6%. *Ralstonia pickettii* was the most commonly identified kind of bacteria isolated from the muscles and the presence was most likely a result of contamination. *Staphylococci* were the second-most commonly found bacteria and were recovered in four out of the 102 carcasses belonging to the case group (~4%), but also in one out of the 60 control animals. The relative risk of finding bacteria in muscles in the case group was five times higher than in the control group (P = 0.01). However, for *Staphylococci* specifically, no difference in risk was found (RR = 0.6; P = 0.86) (Bækbo et al., 2016).

In sows, the muscle samples were in general sterile or below the detection level (97% of the case sows and 95% of the control sows, P = 0.67, implying no difference between the two groups). In the abscesses, the most commonly found bacteria were *Trueperella* spp., found in 66% of the positive samples. *Streptococci* (11%), *Staphylococci* (5%) *E. coli* (5%), and *Pasteurella multocida* (5%) were also found, followed by other bacteria (Pedersen et al., 2017). Kyrval et al. (1995) also found that *Trueperella* spp. were the most common kind of bacteria in abscesses.

Moreover, retrospective data were gathered. For finishing pigs, data from seven large abattoirs were collected. The data covered inspection records from 1 month up to 1 year, lasting from July 2012 to June 2013 and involved a minimum of 456,000 pigs per abattoir. During that period, 40,000 finishing pigs were destined to de-boning due to presence of lesions indicative of a prior septicaemia, corresponding to 0.24%. The proportion of carcasses totally condemned after de-boning was very low at all abattoirs; varying from 0% to 1%. A variation between the seven slaughter plants was observed, indicating that the septicaemia-inspection may not have been undertaken in an equally thorough way at all plants (Bækbo et al., 2016). For sows, the retrospective data covered one large sow abattoir for the period from January 2014 to March 2015. During those 14 months, 322,972 sows were slaughtered and subjected to meat inspection. A total of 1.8% of these were sent to de-boning, and further findings during de-boning were made in 38 sows. Only five carcasses were condemned due to active septicaemia-related lesions. In conclusion, most abscesses were found during the septicaemia-inspection – leaving very few abscesses to be discovered at de-boning.

### 3.6. Comparison of de-boning and targeted inspection of finishing pigs with lesions indicative of prior septicaemia

Based upon the results of the studies presented in Section 3.5, it was hypothesized that a more gentle and targeted procedure for cutting and cleansing of abscesses as part of an own-check program could replace de-boning without jeopardising the ability to detect abscesses. In 2018, a trial was undertaken as a cohort study on two abattoirs slaughtering finishing pigs to assess the safety of such an approach (Petersen & Alban, 2018). The study included 200 pigs per abattoir, yielding a total of 400 pigs with lesions indicative of prior septicaemia. At each abattoir, the 200 pigs were divided into two groups of 100 pigs. A detailed recording of abscesses found during cutting was undertaken. Group 1 pigs were first sent to the re-work area for a septicaemia-inspection and subsequently destined to de-boning, and hence, represented the rules that applied at that time. Group 2 pigs were subjected to the same handling as Group 1 except from that de-boning was replaced with a gentle and targeted procedure for cutting at the rework area. The cutting is

described in the Supplementary materials.

The results showed that the same numbers of abscesses were found during the mandatory septicaemia-inspection in Group 1 and Group 2 (Table 5). During the subsequent cutting/de-boning, no difference in the number of abscesses was found between Groups 1 and 2 within the given abattoir, despite slightly more abscesses were found in the ham/back part in Group 1 compared to Group 2 (10 versus 3;  $P = 0.04$ ) on Abattoir B. Similarly, on Abattoir A, the total number of soft tissue abscesses related to septicaemia were higher (13 versus 4) in Group 1 compared to Group 2 ( $P = 0.02$ ). A higher number of abscesses (29 versus 9) with connection to bones were found on Abattoir B compared to Abattoir A ( $P < 0.001$ ). The conclusion was that the vast part of the abscesses related to prior septicaemia was found in the rework area as part of the septicaemia inspection. Moreover, it was estimated that abscesses can be found in 10–20% of the pigs during the subsequent de-boning and cutting process – irrespective of whether a complete de-boning or a target cutting would be undertaken (Petersen & Alban, 2018). In conclusion, the study demonstrated that targeted cutting was equally good at finding abscesses as complete de-boning. The study led to a possibility of replacing mandatory de-boning of finishing pigs with an own-check program involving targeted cutting into the predilection site for abscesses. The new legislation came into force in September 2018.

### 3.7. Targeted search for abscesses in relation to prior septicaemia in sows

Preliminary studies and analyses using register data from a large sow abattoir in Denmark had shown that almost no abscesses are overlooked at the septicaemia-inspection (Pedersen et al., 2017). However, routine recordings at meat inspection may not have been sufficiently detailed to address this issue. Hence, to elucidate whether and where abscesses could be found after the septicaemia-inspection a prospective study was needed. Based upon the outcome, the examination method could be updated, and the legislation amended.

This cohort study in sows was undertaken in November and December 2018 at the largest Danish sow abattoir (Petersen & Alban, 2019). The study involved 100 sows destined for de-boning, following findings of lesions indicating prior septicaemia. A recording scheme was designed to ensure systematic registration of findings during 1) the septicaemia-inspection, 2) a supplementary examination and 3) de-boning (Fig. 3).

The results show (Fig. 3) that additional abscesses were found at de-boning for seven carcasses. The location was: femur and humerus ( $N = 1$ ), humerus ( $N = 3$ ), hind side of costae (2), and scapula (1). For the abscesses in femur and humerus, the location was latero-proximal to the growth line. Additionally, abscesses were found in seven carcasses in relation to the supplementary examination. Two of these were considered as related to the septicaemia complex; one in the pelvis and one latero-proximal in the humerus. The remaining five abscesses were

found in the neck or the midpart. Hence, abscesses related to septicaemia were found in nine out of 100 carcasses – either in the supplementary examination or the septicaemia-inspection. Moreover, neck abscesses related to injections were found in 54 out of 100 carcasses.

The study led to an update of the official septicaemia-inspection (See Figs. S1 and S2) to be done on the hanging and split sow carcass (implying improved working conditions), and to a possibility of replacing mandatory de-boning of sows with the updated septicaemia examination. Fig. 4 shows the difference between the original septicaemia-inspection and the additional cuts, which can be made as a supplementary examination by the abattoir as part of an own-check program. The new legislation came into force in April 2019.

## 4. Discussion

As stated in the introduction, the adoption of the EU Meat inspection Regulation 854/2004 made it possible to implement changes to meat inspection of pigs raised under controlled housing conditions, pending that a risk assessment could show that such a change would not be associated with an increased risk for human or animal health or welfare (Anonymous, 2004). The focus in the work undertaken in the different countries – both before and after 2004 – was mainly to illustrate the effect of replacing traditional meat inspection with VOI (see e.g. Hamilton et al., 2002; Mousing et al., 1999, 1997; Nesbakken, 2006). This was due to the hypotheses that palpations and incisions are mostly used to detect lesions caused by microorganisms that are irrelevant for public health and that abscesses are mainly aesthetic meat quality issues that do not affect food safety – as explained by Laukkanen-Ninios et al. (2020). Moreover, in 2011 an EFSA Opinion concluded that VOI of swine would lead to lower probability of between-carcass or between-organ spread of food safety hazards such as *Salmonella* spp. due to less handling and manipulation (EFSA, 2011a). In the EFSA Opinion, the age and the way of raising the pigs was not seen as a constraint to this conclusion (EFSA, 2011a). But the EFSA Opinion did not go into details of what would be missed and what the impact would be regarding food safety, animal health, and animal welfare. Neither did the EFSA Opinion go into the discussion of how to handle cases with lesions indicative of prior septicaemia, a question raised by Alban et al. (2011).

In 2014, the EU Meat Inspection Regulation was adapted, and the current legislation stipulates that meat inspection of swine should be visual-only. An exception is warranted if information revealed during ante- or post-mortem or from the herd of origin points to a need for traditional inspection involving incisions and palpations. Moreover, it is a requirement that food chain information is sent to the abattoir prior to the slaughter of the animals (Anonymous, 2014). EU Regulations are supposed to come into force in all EU Member States on the date stipulated in the regulation. For several Member States, national implementation of the new meat inspection regulation was hampered due to

**Table 5**

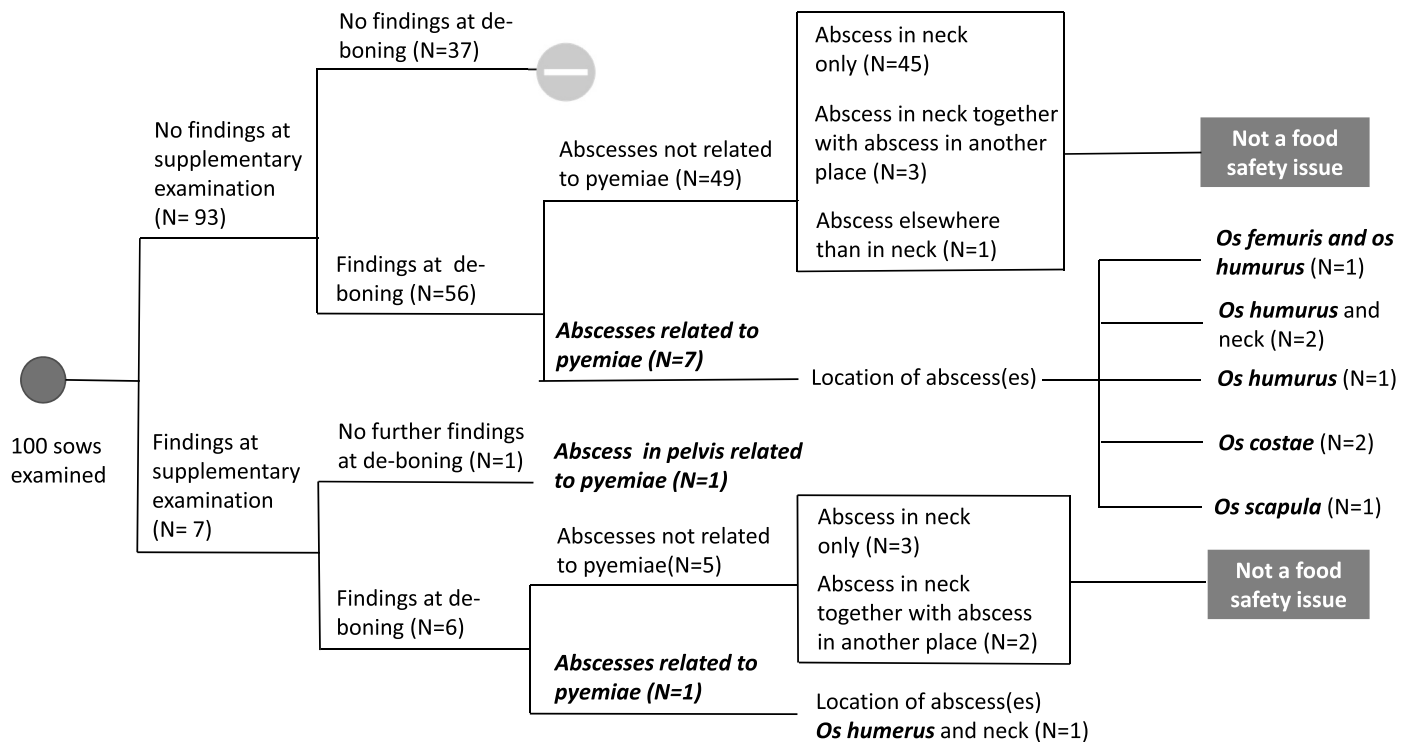
Results of an implementation study undertaken in two Danish abattoirs, involving 400 finishing pigs with lesions indicative of prior septicaemia, subjected to mandatory septicaemia inspection followed by de-boning (Group 1) or targeted cutting (Group 2), 2018.

| Abattoir | Group (No. of pigs) | Abscesses <sup>a</sup> found in pig during septicaemia inspection |      |         |             | Abscesses <sup>a</sup> found in pig during subsequent cutting/de-boning |            |             |             |                 |                 |  | Total no. of pigs with abscess(es) in |    |
|----------|---------------------|-------------------------------------------------------------------|------|---------|-------------|-------------------------------------------------------------------------|------------|-------------|-------------|-----------------|-----------------|--|---------------------------------------|----|
|          |                     | Column                                                            | Ribs | Sternum | Other bones | Neck                                                                    | Front part | Middle part | Tender-loin | Ham/hind part   | Other locations |  |                                       |    |
| A        | 1 (100)             | 42                                                                | 34   | 16      | 16          | 0                                                                       | 7          | 2           | 0           | 2               | 5               |  | 13 <sup>b</sup>                       | 6  |
|          | 2 (100)             | 49                                                                | 40   | 15      | 26          | 0                                                                       | 2          | 1           | 0           | 2               | 5               |  | 4 <sup>b</sup>                        | 3  |
| B        | 1 (100)             | 39                                                                | 33   | 9       | 23          | 0                                                                       | 4          | 3           | 0           | 10 <sup>c</sup> | 4               |  | 12                                    | 17 |
|          | 2 (100)             | 30                                                                | 40   | 13      | 21          | 2                                                                       | 10         | 4           | 0           | 3 <sup>c</sup>  | 1               |  | 10                                    | 12 |

<sup>a</sup> All these abscesses were judged by the meat inspectors to be part of the septicaemia complex.

<sup>b</sup> P-value = 0.02.

<sup>c</sup> P-Value = 0.04.



**Fig. 3.** Distribution of lesions found in 100 sows destined for de-boning, following findings of lesions indicating prior septicaemia. A recording scheme was designed to ensure systematic registration of findings during a 1) supplementary examination after the mandatory septicaemia-inspection and 2) a subsequent de-boning.

various reasons: 1) reporting of food chain information was not fully put in place, 2) some meat inspectors were concerned about the safety of the VOI, as also shown by [Laukkanen-Ninios et al. \(2020\)](#), or 3) the Member State had an export out of the EU to third countries, which would not necessarily consider VOI as equivalent to their own kind of meat inspection ([Bækbo et al., 2015](#)).

Today, the meat inspection of Danish finishing pigs reared under controlled housing conditions is done in accordance with the EU Regulation. However, for non-controlled pigs slaughtered in the export-authorized abattoirs, traditional inspection takes place due to export requirements. For a full overview of the challenges, opportunities and lessons learned related to implementation of visual-only swine inspection in EU, please see [Riess & Hoelzer, \(Under revision\)](#). Future negotiations with trade partners may involve the question of whether it is necessary to undertake traditional meat inspection for outdoor-reared pigs, as discussed by [Hansen et al. \(2018\)](#). Recently, the Food and Agricultural Organisation (FAO) of the United Nations has published a technical guidance regarding risk-based meat inspection ([FAO, 2019](#)). This shows that the modernisation process is not just related to the EU.

Our work began several years before the EFSA Opinion was published in 2011 and continued several years after. During those years, we looked in detail into the issues not fully covered by EFSA, such as what would happen if routine incisions or palpations of specific organs were omitted. To address these issues, we applied a risk assessment approach using in-country-up-to-date data to provide a science-based and transparent answer. We used the OIE approach and not the Codex alimentarius approach. The difference between the two frameworks is minor and primarily relates to the order of the elements ([Codex alimentarius, 2014](#); [OIE, 2010](#)).

Our work has been in line with the EFSA Opinion's recommendation of replacing traditional inspection with VOI for finishing pig carcasses, which appear normal at *ante*- and *post-mortem* inspection, and where the food chain information does not indicate anything abnormal. The gradual approach used in our work ensured proper communication with

several groups of stakeholders: the meat inspectors, the veterinary authorities, the meat industry, and the trade partners. The main limitation of our work is related to the number of samples taken; in general, the numbers were low because of budget constraints. Moreover, in some projects we only collected samples from one abattoir during some weeks or a few months but not an entire year. To compensate for this, we collected retrospective data from the different available registers, such as the Danish meat inspection database, as well as compared our findings with those in the literature. Therefore, we believe that our findings are representative of the typical Danish conditions.

The hypothesis is that the contamination rate of the carcasses would be lower if VOI is replacing traditional meat inspection ([Laukkanen-Ninios et al., 2020](#)). However, one may assume that the role of the meat inspector may be limited compared to the overall handling of the carcasses. Therefore, the conclusion may be that both inspection systems are likely to result in the same level of consumer protection – as stated by [Hamilton et al. \(2002\)](#). We did not focus on the contamination rates – we were more interested in understanding what would be missed, how often, and what the consequences would be of missing a lesion.

The EFSA Opinion about hazards to be covered by meat inspection of swine stated that meat inspection needs to be re-evaluated at regular intervals to ensure that it is targeting the most relevant hazards ([EFSA, 2011a](#)). Such a re-evaluation is in fact what we have done. For example, the studies into the effect of de-boning of animals with lesions indicative of prior septicaemia helped to update the way the extended septicaemia-inspection is taking place for these animals at the rework area. This opened for a replacement of the mandatory de-boning with the new extended inspection ([Anonymous, 2019b](#)) as well as an own-check program involving a targeted examination leading to a €3 million reduction in the annual costs and less food waste without jeopardising food safety, animal health, or welfare.

The EFSA Opinion from 2011 recommended a transition from traditional meat inspection involving macroscopic evaluation of carcasses and organs to development of comprehensive meat safety



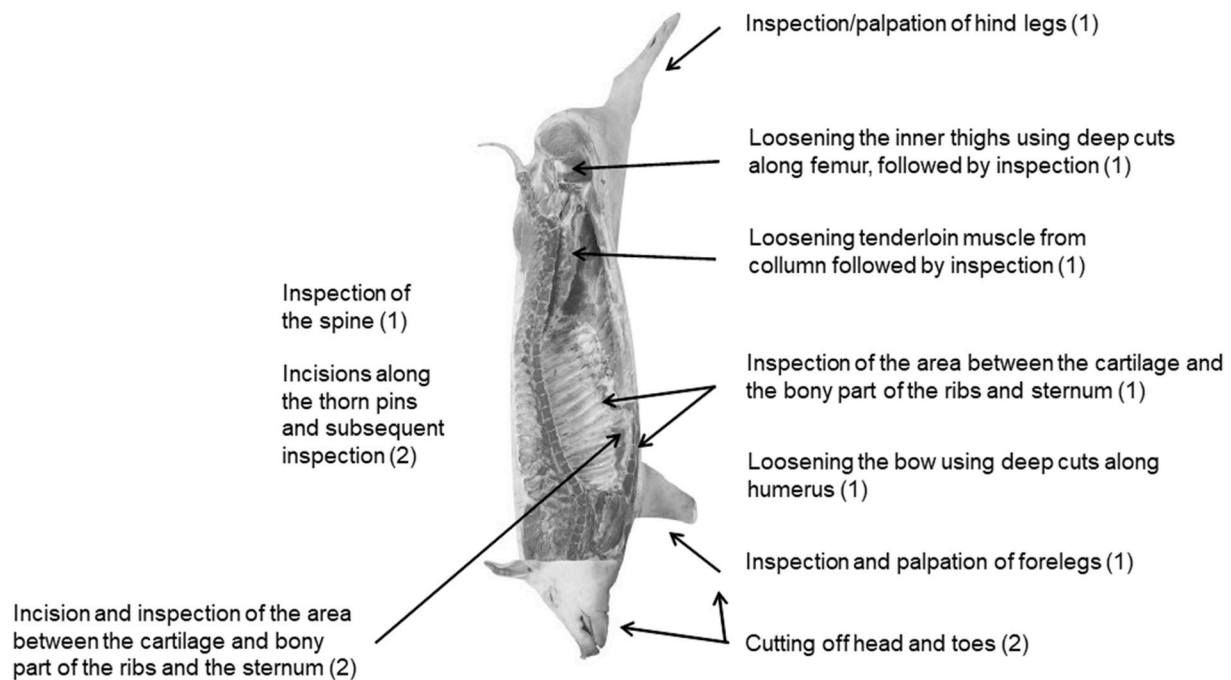


Fig. 4. Graphical description of the septicaemia-inspection (1) and the supplementary examination (2) consisting of inspection, palpation and incision at defined locations at the carcass.

assurance systems based upon risk and with a view of the entire supply chain (EFSA, 2011 ab). This should be based upon the hazards identified as relevant, which according to the EFSA Opinion for swine is *Salmonella*, *Yersinia*, *Trichinella* and *Toxoplasma* (EFSA, 2011a). A similar report from another working group within EFSA, published at the same time, contains a technical specification on so-called harmonized epidemiological indicators (HEIs) for public health hazards to be covered by meat inspection of swine (EFSA, 2011b). The list of relevant hazards for pigs includes mycobacteria and *Taenia solium* as well as the four hazards, listed in the EFSA Opinion and for which a formal risk-ranking approach was used (EFSA, 2011a). The identification of mycobacteria as a relevant hazard for meat inspection of swine is peculiar, as there is currently no evidence of pork-related transmission of mycobacteria to humans. Human infection is known to occur, for example, from unpasteurised milk or as a result of direct contact/inhalation of aerosols from infected animals (EFSA, 2011a; O'Reilly & Daborn, 1995). Hence, mycobacteria represent an occupational hazard in an abattoir. Moreover, *Mycobacterium bovis* is a serious animal health issue, and maybe therefore, detection of suspect lesions in pigs during meat inspection is suggested as a harmonized epidemiological indicator (EFSA, 2011b).

In Denmark, *Salmonella* is covered in a surveillance programme in finishing pigs (Alban et al., 2012). Moreover, *Trichinella* is not found in the indoor compartment and surveillance can legally be replaced by auditing of biosecurity by a third-party independent auditor – as explained by Alban and Petersen (2016). *Yersinia* is supposedly dealt with through focus on hygiene during slaughter, and for *Toxoplasma*, a pilot study has shown a very low frequency in the indoor finishing pig compartment, medium in the outdoor finishing pig compartment and high in the sow compartment (Kofoed et al., 2017). Surveillance for *Toxoplasma* is not required by the EU, but auditing of biosecurity in the low-risk compartment as well as risk-mitigation without surveillance could be considered in the high-risk compartment (Alban et al., 2020). Denmark has not had cases of bovine tuberculosis for more than 30 years, and infection has not been found in wildlife. Hansen et al. (2018) evaluated the effect of VOI and traditional meat inspection of outdoor-reared pigs in Denmark. Infection will be detected later with VOI compared to traditional inspection, but it will not have spread much, because the movement of pigs in Denmark is by contract within

closed circles. Therefore, it may be more cost-effective to focus on imported high-risk animals coming into contact with Danish livestock rather than using traditional inspection on all pigs from non-controlled housing (Hansen et al., 2018). Hill et al. (2013) also concluded that the overall risk from all pig-related hazards to public health associated with VOI replacing traditional inspection was assessed as negligible for all pigs, irrespective of rearing system. *Taenia solium* has not been observed in Denmark for decades, and the year of the last case is unknown (Anonymous, 2019a).

Recently, Felin (2019) and Felin et al. (2019) studied the application of meat safety assurance systems in Finnish pig production and concluded that such systems are feasible. However, the question is not just whether meat safety assurance systems are feasible but also how they can become useful for those who will have to pay for such systems. To study this in detail, a so-called COST Action Network about risk-based meat inspection and integrated meat safety assurance (RIBMINS) has been established. RIBMINS is currently actively involving more than 100 persons from more than 30 mainly European countries, in modernisation of meat inspection. The activities include illustration of the effect of various diagnostic technologies such as camera-based image technology called “vision”, use of meat-juice serology for on-farm classification (Felin et al., 2019; Meemken et al., 2014) or acute-phase proteins (Blagojevic et al., 2011). Decontamination using hot water (Alban & Sørensen, 2010), steam, or organic acids as barriers to implementation of meat safety assurance systems will also be covered. More can be read about RIBMINS on: <https://ribmins.com/>.

## 5. Conclusion

Worldwide, meat inspection is undergoing a change to ensure further focus on food safety risks of animal origin. The Danish experience is that risk assessment following the OIE approach combined with using in-country up-to-date data has shown to be a transparent approach to illustrate what the effect of a given change may be on food safety, animal health, and animal welfare. Transparency ensures acceptance of equivalence on export markets as well as among consumers and people involved in meat inspection. Hereby, targeted, cost-effective and safe procedures may be implemented step-by-step, while also ensuring a

reduction in food waste and improving the working environment. Our work has shown that routine incisions and palpations are not needed as part of inspection of finishing pigs raised indoors, unless the pigs show visually recognisable lesions. Moreover, for lesions related to prior septicaemia, targeted focus on the predilection sites for abscesses in finishing pigs and sows during the inspection at the rework area and during cutting is sufficient to detect and handle abscesses.

## Declaration of competing interest

The authors LA, JVP, and AKB work for an organization that gives advice to farmers and abattoirs, and GP works for the Danish Veterinary and Food Administration as an official veterinarian.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodcont.2020.107450>.

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